

Video Article

Efficient Synthesis of All-Carbon Quaternary Centers via the Conjugate Addition of Functionalized Monoorganozinc Bromides

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Abstract

The conjugate addition of organometallic reagents to α,β -unsaturated carbonyls represents an important method to generate C–C bonds in the preparation of all-carbon quaternary centers. Though conjugate additions of organometallic reagents are typically performed utilizing highly reactive organolithium or Grignard reagents, organozinc reagents have garnered attention for their enhanced chemoselectivity and mild reactivity. Despite numerous recent advances with more reactive diorganozinc and mixed diorganozinc reagents, the generation of all-carbon quaternary centers via the conjugate addition of functionalized monoorganozinc reagents remains a challenge. This protocol details a convenient and mild “one-pot” preparation and copper mediated conjugate addition of functionalized monoorganozinc bromides to cyclic α,β -unsaturated carbonyls to afford a broad scope of all-carbon quaternary centers in generally excellent yield and diastereoselectivity. Key to the development of this technology is the utilization of DMA as a reaction solvent with TMSI as a Lewis acid. Notable advantages to this methodology include the operational simplicity of the organozinc reagent preparation afforded by the utilization of DMA as a solvent, as well as an efficient conjugate addition mediated by various Cu(I) and Cu(II) salts. Moreover, an intermediate silyl enol ether can be isolated utilizing a modified workup procedure. The substrate scope is limited to cyclic unsaturated ketones, and the conjugate addition is impeded by stabilized (e.g., allyl, enolate, homoenolate) and sterically encumbered (e.g., neopentyl, *o*-aryl) monoorganozinc reagents. Conjugate additions to five- and seven-membered rings were effective, albeit in lower yields compared with six-membered ring substrates.

Video Link

The video component of this article can be found at <https://www.jove.com/video/59775/>

Introduction

The formation of carbon-carbon bonds is arguably the most important and powerful transformation in organic chemistry. The conjugate addition of organometallic reagents to α,β -unsaturated carbonyls comprises one of the most versatile methods for the construction of C–C bonds, especially in the challenging generation of all-carbon quaternary centers^{1,2}. Despite the central importance of the conjugate addition of organometallic reagents to the formation of quaternary centers, few methodologies address the challenge of incorporating sensitive functional groups in these reactions. Indeed, in the majority of these transformations, highly reactive organolithium, Grignard, or diorganozinc reagents are the nucleophiles of choice. These reactive organometallics, however, are incompatible with many sensitive functional groups, thereby limiting the complexity of both the α,β -unsaturated carbonyl and organometallic reagent, often necessitating the use of protecting groups or alternative strategies in multi-step synthesis.

Monoorganozinc reagents are an attractive class of organometallic reagents which have garnered widespread attention for their mild reactivity and enhanced functional group compatibility^{3,4,5,6}. In spite of their exceptional functional group tolerance and trivial preparation from organohalides, there are few examples of monoorganozinc reagents in the conjugate addition to β,β -disubstituted α,β -unsaturated carbonyls to generate quaternary centers^{7,8,9}. Furthermore, these transformations typically require stoichiometric quantities of toxic cyanocuprate reagents with one report demonstrating minimal catalyst turnover^{10,11,12,13}. The objective of our study is to establish a simple and practical catalytic method for the conjugate addition of functionalized monoorganozinc reagents to α,β -unsaturated carbonyls to generate all-carbon quaternary centers. Toward this end, we have developed a protocol utilizing *N,N*-dimethylacetamide (DMA) as a solvent with chlorotrimethylsilane (TMSI) as a Lewis acid which enables a “one-pot” copper catalyzed (20 mol %) conjugate addition of functionalized monoorganozinc reagents to α,β -unsaturated carbonyls to generate a broad scope of all-carbon quaternary centers in high yield¹⁴.

The utilization of DMA as a solvent has several notable advantages over methods reported in the literature. DMA improves the efficiency of zinc insertion into organohalides which obviates the requirement for expensive and hygroscopic additives such as LiCl employed in ethereal solvent systems¹⁵. This also expands the scope of direct zinc insertion from sensitive, often commercially unavailable organoiodides to more stable and widely accessible organobromides¹⁶. The protocol detailed herein generates alkyl monoorganozinc reagents (**2**) from diverse organobromides, which are used in situ in the formation of a reactive cuprate complex that engages cyclic α,β -unsaturated ketones in a conjugate addition reaction (**Figure 1**). DMA also enables the reaction to proceed with cheaper and less toxic copper sources such as CuBr·DMS, eliminating stoichiometric

toxic waste generated with CuCN utilized in other reports^{10,11,12,13}. Our standard reaction conditions provide access to a broad scope of β -quaternary ketones (**5**) with both five-, six-, and seven-membered ring conjugate acceptors obtained via the hydrolysis of an intermediate silyl enol ether (**4**). The intermediate silyl enol ether was observed to be moderately stable and could be isolated in excellent yield utilizing a modified workup procedure.

Protocol

CAUTION: Consult Material Safety Data Sheets (MSDS) prior to the use of the chemicals in this procedure. Use appropriate personal protective equipment (PPE), including safety glasses, a lab coat, and nitrile or butyl gloves as many of the reagents and solvents are corrosive, toxic, or flammable. Carry out all reactions in a fume hood. It is necessary to flame-dry glassware and use an inert atmosphere (nitrogen or argon) for this protocol. Liquids used in the first two steps of the protocol are syringe transferred.

1. Functionalized monoorganozinc bromide formation¹⁶

1. Add zinc dust (0.8173 g, 12.50 mmol, 2.5 equiv), DMA (6.0 mL), and I₂ (126.9 mg, 0.500 mmol, 0.10 equiv) to a flame-dried 50 mL Schlenk reaction flask containing a stir bar under argon. Stir the suspension at ambient temperature until the brownish-orange color completely dissipates to a gray suspension.

CAUTION: DMA is flammable, acutely toxic, and a mild irritant.

2. Add ethyl 4-bromobutyrate (**6**; 1.43 mL, 10.0 mmol, 2.0 equiv) to the gray suspension. Immerse the flask in an 80 °C oil bath with vigorous stirring until consumption of the organobromide is observed by gas chromatography (GC) analysis.

CAUTION: Ethyl 4-bromobutyrate is flammable and a mild irritant.

1. Briefly dip a disposable glass Pasteur pipet into the reaction mixture, remove from the flask and rinse the aliquot with ca. 0.5 mL of diethyl ether (Et₂O) into a 2 mL vial containing ca. 0.5 mL of saturated NH₄Cl. Vigorously shake the vial and analyze the organic layer by GC.

NOTE: Reaction progress is determined by comparing the disappearance of the organobromide peak to the appearance of the protonated organozinc reagent, which typically has a shorter retention time.

CAUTION: Et₂O is flammable and acutely toxic.

3. Cool the monoorganozinc reagent to ambient temperature. The remainder of this conjugate addition protocol can be paused at this stage for several hours to overnight under inert atmosphere with minimal impact on the yield of the conjugate addition product.

2. Monoorganozinc bromide conjugate addition to α,β -unsaturated ketones

1. Cool the organozinc suspension in an ice/H₂O bath for ca. 5 min, and then add CuBr·DMS (205.6 mg, 1.000 mmol, 0.20 equiv) and additional DMA (10.5 mL, 0.30 M overall with respect to unsaturated ketone), and stir the reaction for ca. 10 min.

2. Add TMSCI (1.52 mL, 12.0 mmol, 2.4 equiv) to the cooled suspension, followed by 3-methyl-2-cyclohexenone (**13**; 567 μ L, 5.00 mmol, 1.0 equiv). Remove the cooling bath after ca. 30 min and monitor the reaction until the α,β -unsaturated ketone is consumed by TLC analysis, or up to 24 h.

NOTE: TLC analysis is performed with silica-coated glass plates and developed with 4:1 hexanes–ethyl acetate. *R_f* values: unsaturated ketone **13** = 0.15; silyl enol ether **37** = 0.61; ketone product **21** = 0.21. Visualization is achieved by UV-quenching at 254 nm, followed by *p*-anisaldehyde staining.

CAUTION: TMSCI is flammable, acutely toxic, a skin irritant, and corrosive. 3-methyl-2-cyclohexenone is flammable and acutely toxic.

3. Extraction work-up and purification

1. Add acetic acid (1.5 mL, ca. 5 equiv) to the completed conjugate addition reaction to hydrolyze the intermediate silyl enol ether into the ketone product. Monitor the progress of hydrolysis in ca. 15 min intervals by TLC analysis using conditions from step 2.2.

CAUTION: Acetic acid is flammable and corrosive.

1. In the event that the silyl enol ether remains after 1 h, add tetrabutylammonium fluoride (TBAF; 1 M solution in THF, 0.5–1.0 equiv at a time) to facilitate complete hydrolysis as evident by TLC.

2. Add 1 M HCl (15 mL) to the reaction flask, mix well, and then transfer the reaction contents into a 250 mL separatory funnel. Rinse the flask with Et₂O (20 mL) and H₂O (15 mL), adding the rinses to the separatory funnel. Gently shake the contents of the funnel, venting between each mixing, and allow the layers to separate. Drain the bottom aqueous layer into a 125 mL Erlenmeyer flask, and then drain the organic layer into a separate 250 mL Erlenmeyer flask.

3. Return the aqueous layer to the separatory funnel and extract with four separate portions of Et₂O (4 x 30 mL), adding each organic extraction to the organic-containing Erlenmeyer flask.

4. Add the combined organic extractions to the separatory funnel and sequentially wash with saturated aqueous NaHCO₃ (25 mL), then saturated aqueous NaCl (25 mL). Drain each aqueous washing into the aqueous-containing Erlenmeyer flask, and drain the final organic layer into a dry 250 mL Erlenmeyer flask.

5. Dry the organic layer over MgSO₄ and vacuum filter into a 250 mL round bottom flask using a glass-fritted Buchner funnel. Rinse the solids on the frit with a small portion of additional Et₂O.

6. Concentrate the filtrate under reduced pressure using a rotary evaporator. Place the flask with remaining residue under high vacuum (ca. 75–200 mtorr) for at least 10 min. Analyze a sample of the crude residue by ¹H NMR using CDCl₃.

7. Purify the crude oil by automated flash chromatography using a silica gel (SiO₂) stationary phase with a dry-loaded sample and elute with ethyl acetate in hexanes.

1. Dry-load the sample by dissolving the crude oil in a minimal amount of Et₂O, and then transfer this solution to a prepacked SiO₂ cartridge (25 g). Apply a reduced pressure to the bottom of the loading column for ca. 5 min to remove the excess solvent.
2. Elute the sample using a prepacked SiO₂ column (100 g) with a gradient of ethyl acetate in hexanes (5% → 25%), collecting the column effluent in test tubes.
3. Assay fraction purity using TLC analysis (conditions in Step 2.2). Combine and rinse all fractions containing the desired quaternary ketone into a tared round bottom flask.
8. Concentrate the solution under reduced pressure on a rotary evaporator and remove the final volatiles under high vacuum for at least 30 min. Obtain a final mass of the flask and analyze a sample of the purified product by ¹H NMR using CDCl₃. CAUTION: Hexanes and ethyl acetate are flammable. SiO₂ powder is a respiratory irritant.

Representative Results

Conjugate addition product ethyl 4-(1-methyl-3-oxocyclohexyl)butanoate (**21**) was isolated as a clear, colorless oil (1.0372 g, 4.583 mmol, 92% yield) using this efficient one-pot protocol. ¹H and ¹³C NMR spectra are presented in **Figure 2** and **Figure 3** to confirm the structure and purity. Of specific note in the ¹H spectrum analysis is the presence of a two proton AB quartet at δ 2.15 ppm, indicating that the diastereotopic C2 hydrogens spin couple. A three-proton singlet at δ 0.94 ppm represents the C1 quaternary methyl group.

Ethyl 4-(1-methyl-3-oxocyclohexyl)butanoate (21**).** *R_f* = 0.21 (4:1 hexanes/ethyl acetate); ¹H NMR (400 MHz, CDCl₃): δ 4.13 (q, *J* = 7.1 Hz, 2H), 2.32–2.23 (m, 4H), 2.15 (ABq, Δδ_{AB} = 0.06, *J*_{AB} = 13.5 Hz, 2H), 1.87 (quintet, *J* = 6.4 Hz, 2H), 1.60 (dt, *J* = 23.3, 15.4, 7.6 Hz, 4H), 1.34–1.21 (m, 2H), 1.26 (t, *J* = 7.1 Hz, 3H), 0.94 (s, 3H). ¹³C{¹H} NMR (101 MHz, CDCl₃): δ 211.3, 172.9, 59.9, 53.4, 40.7, 40.6, 38.2, 35.3, 34.3, 24.7, 21.8, 18.7, 14.0; IR (ATR): 2951, 2939, 2873, 1730, 1708, 1374, 1178, 1025 cm⁻¹; HRMS (DART+) *m/z*: [M + H]⁺ calcd for C₁₃H₂₃O₃ 227.1642, found 227.1640.

A collection of cyclic ketone addition products with β-quaternary centers were prepared in good to excellent yields using this simple and efficient one-pot protocol (**Figure 4**)¹⁴. All reaction products were analyzed by ¹H and ¹³C NMR, as well as high-resolution mass spectrometry (HRMS), and found to be of high purity. In addition to the incorporation of ester (**21**), nitrogen (**22**, **24**), and halide (**23**, **27**) functionality, this reaction protocol furnishes products with various ring sizes (**29–31**) and high levels of stereoselectivity ((±)-**32**–(±)-**35**) when using chiral α,β-unsaturated ketones ((±)-**17**–(±)-**20**). Diastereomer ratios are determined by routine integration of select peaks in the product ¹H spectrum with the major diastereomer shown¹⁴. It is evident from these examples that the favored pathway involves delivery of the organic fragment to the alkene face opposite to non-H groups at the γ- and δ-position of the α,β-unsaturated ketone.

Monoorganozinc formation on this scale generally requires heating at 80 °C for 2.5–3 h and yields a colorless to pale-yellow suspension with zinc dust remaining. GC analysis provides an ideal method for the detection of organobromide reagents during this step owing to their challenging visualization by TLC analysis. Progress of the conjugate addition (Step 2.2) and enol ether hydrolysis (Step 3.1) are monitored by TLC analysis. Unsaturated ketone **13** (*R_f* = 0.15, 4:1 hexanes-ethyl acetate) is UV-active and stains, whereas the intermediate silyl enol ether (*R_f* = 0.61) and ketone product **21** (*R_f* = 0.21) stain only. Incomplete hydrolysis of the enol ether is generally indicated by the presence of a singlet at δ 4.64 in the ¹H NMR spectrum (Step 3.6). Standard flash chromatography is adequate for the purification of conjugate addition products¹⁷.

Optimal yields of the conjugate addition products are obtained when using the established reagent equivalence for the monoorganozinc (2.0), copper catalyst (20 mol %), and Lewis acid (2.4) parameters. A diminished 72% yield of **21** was observed with a decrease in CuBr·DMS to 10 mol% (**Table 1**, entries 1–3). Reducing the monoorganozinc bromide (**36**) or TMSCl equivalence to 1.2 also resulted in a modest decrease in the production of **21** (entries 4 and 5). Notably, the conjugate addition does not proceed without CuBr·DMS or TMSCl (entries 3 and 6).

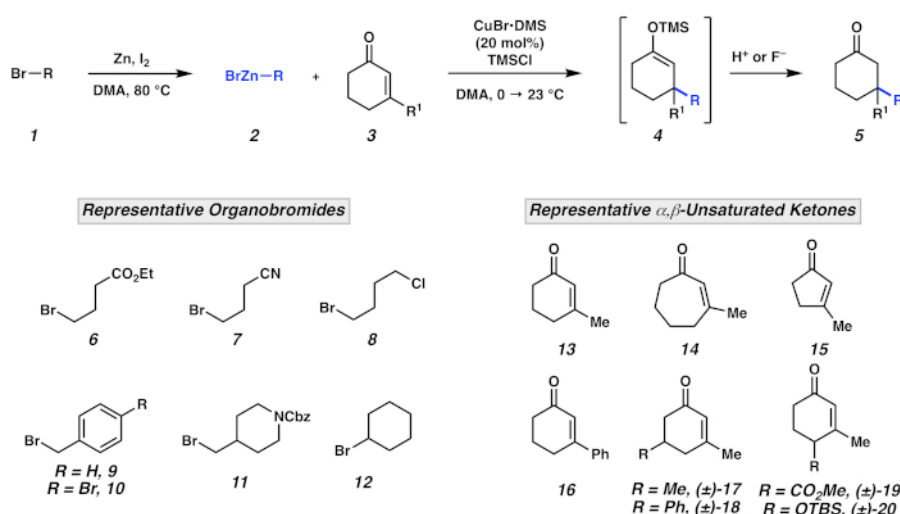


Figure 1. General scheme for a one-pot conjugate addition protocol. Reaction overview with representative substrates. [Please click here to view a larger version of this figure.](#)

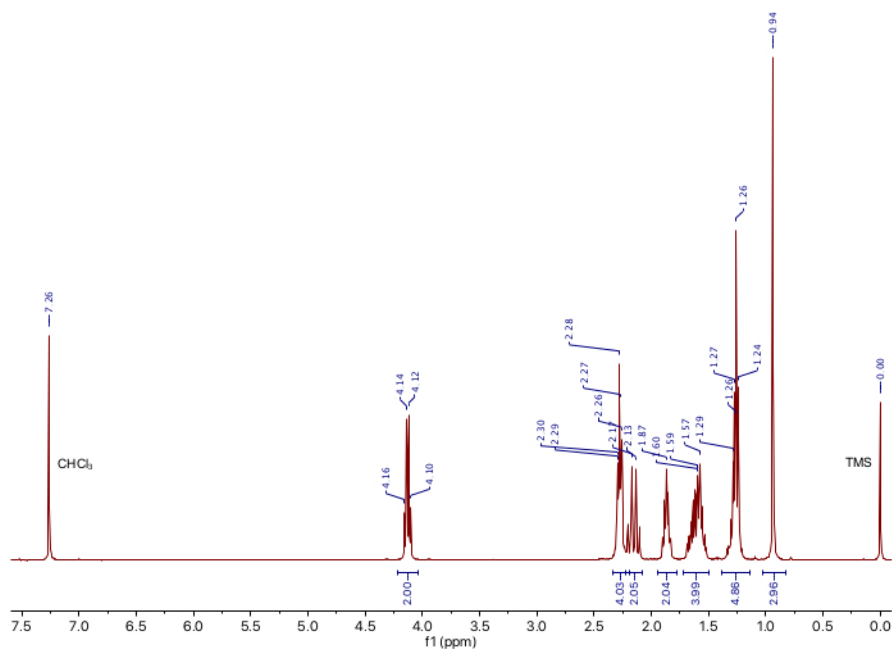


Figure 2. ^1H NMR of 21. Spectrum obtained in CDCl_3 at 400 MHz. [Please click here to view a larger version of this figure.](#)

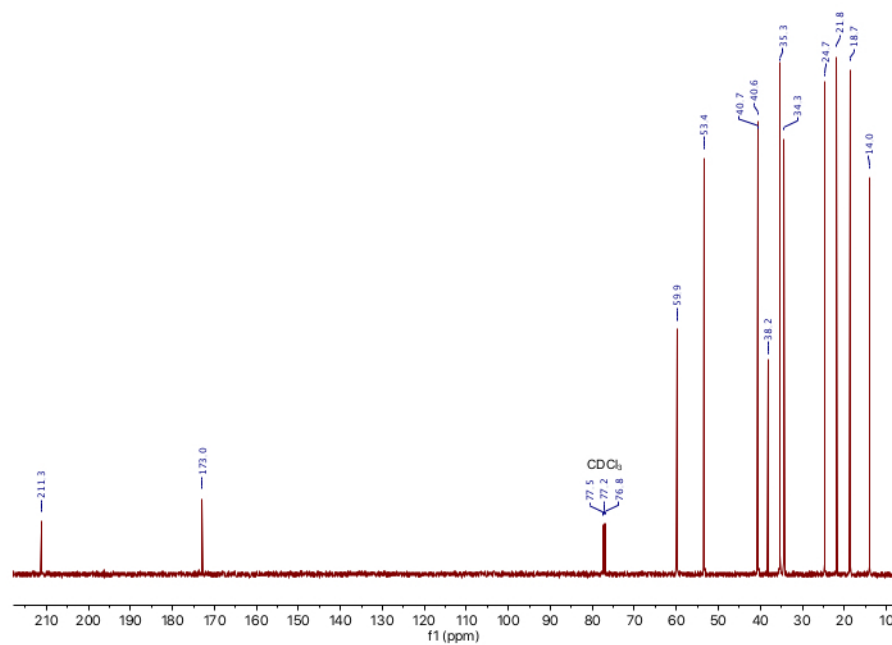


Figure 3. $^{13}\text{C}\{^1\text{H}\}$ NMR of 21. ^1H -decoupled spectrum obtained in CDCl_3 at 101 MHz. [Please click here to view a larger version of this figure.](#)

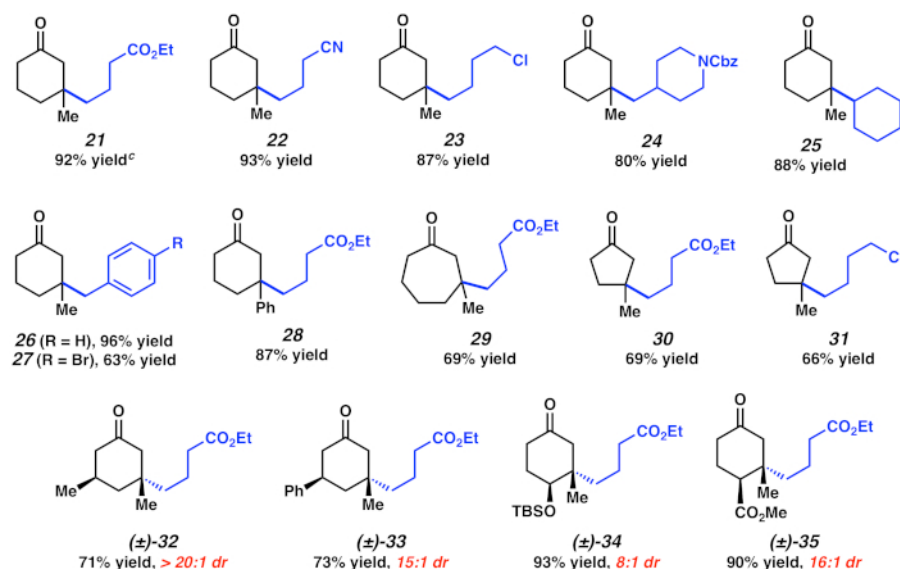


Figure 4. Representative reaction scope for the one-pot conjugate addition of functionalized monoorganozincs. Yields indicate isolated analytically pure products conducted on 1.00 mmol scale of unsaturated ketone. Product **21** was isolated on 5.00 mmol scale of unsaturated ketone. [Please click here to view a larger version of this figure.](#)

Reaction scheme showing the conjugate addition of monoorganozinc 36 (in situ) to enone 13 to form product 21 via intermediate 37. Reagents: CuBr·DMS, TMSCl, DMA, 0 → 23 °C. Workup: AcOH.				
entry ^a	36 (equiv)	CuBr·DMS (mol %)	TMSCl (equiv)	yield 21 (%) ^b
1	2	20	2.4	90
2	2	10	2.4	72
3	2	0	2.4	0 ^c
4	1.2	20	2.4	72 ^c
5	2	20	1.2	73 ^c
6	2	20	0	0
7	2	20	2.4	0 (74) ^d

^a 1.00 mmol **13** according to standard protocol. ^b Isolated yields. ^c Incomplete conversion of **13**. ^d Yield in parenthesis of **37**, obtained from a NaHCO₃ workup that omitted AcOH and HCl.

Table 1. Parameter optimization of the one-pot conjugate addition. A survey of reagent equivalence and reaction outcome.

Discussion

The method detailed herein was developed to harness mild functionalized monoorganozinc reagents in a simple and efficient conjugate addition reaction for the synthesis of β-quaternary ketones¹⁴. Excellent yields and significantly improved catalyst efficiencies were observed through the use of the polar, aprotic solvent DMA with TMSCl. Monoorganozinc formation is aided by DMA, facilitating the direct zinc insertion into readily available alkyl bromides to create a broad scope of functionalized reagents¹⁶.

Monoorganozinc bromide formation is the most important step in this protocol. We have obtained modest to excellent success for conjugate addition reactions when the conversion to the organozinc reagent is quantitative, given the proper equivalence of catalyst and TMSCl. Organobromides with increased substitution near the C-Br bond undergo zinc insertion at a diminished rate compared to the typical 3 h time frame, including substrate **11** (28 h) and substrate **12** (80 h). However, stabilized benzyl organozinc reagents from **9** and **10** are formed at an increased rate (ca. 1 h) at 21–40 °C. The corresponding functionalized cuprate reagents perform well in conjugate addition reactions (**24**, **25**, **26**, and **27**, **Figure 4**) despite this temperature difference.

Upon successful conjugate addition, the silyl enol ether intermediate is hydrolyzed with acid to afford the final ketone product. Acetic acid was the optimal organic-soluble Brønsted acid to facilitate this mild hydrolysis step when compared to aqueous HCl. Silyl enol ethers that are recalcitrant

to hydrolysis can be readily cleaved to ketone products with TBAF. Notably, the synthetically useful¹⁸ silyl enol intermediate **37** could be isolated in 74% yield using this one-pot protocol by substituting the acid workup with NaHCO₃ (Table 1, entry 7).

Challenging purifications can result from incomplete conversion of the unsaturated ketone, owing to the similar *R_f* values of the substrate and ketone product. The modest volatility of the majority of unsaturated ketones used in this study enable their removal through prolonged exposure (> 1 h) to high vacuum.

The broad scope of organobromides and unsaturated ketones for this one-pot protocol yields an array of functionalized β-quaternary ketones¹⁴. This method is currently limited to cyclic unsaturated ketones, as we have not explored acyclic substrates. Quaternary ketones have been generated from unsaturated five- and seven-membered rings, albeit in modest yield⁸ presumably due to differences in overlap of the conjugated system. The conjugate addition also does not proceed with several stabilized (e.g., allyl, enolate, homoenolate) and sterically encumbered (e.g., neopentyl, o-aryl) monoorganozinc reagents¹⁴. We are currently investigating the reactivity of monoorganozinc reagents with respect to Lewis acid and Lewis base additives, in an effort to improve the reaction efficiency and incorporate a more significant scope of organobromides and unsaturated ketones. Excellent yields and selectivities are maintained with a marked improvement in catalyst efficiency when using monoorganozinc reagents that have been filtered from zinc solids after insertion¹⁴. Moreover, various Cu(I) and Cu(II) salts facilitate efficacious conjugate additions when coupled with filtered monoorganozinc reagents.

In summary, we have described an efficient one-pot protocol for the conjugate addition of functionalized monoorganozinc reagents in the preparation of functionalized β-quaternary ketones. This protocol comprises a convenient method to explore the feasibility of various organobromide and unsaturated carbonyl combinations (Figure 1) toward these valuable products.

Disclosures

The authors have no competing financial interests.

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